



GAMEWATCH

PLAY • MANAGE • CONNECT

The All-in-One PS Management System
for Staff, Players & Customers



PLAY
FOR PLAYERS



MANAGE
FOR STAFF



CONNECT
FOR CUSTOMERS

Master Product Requirements Document

PRD v1.0 — Product Blueprint

Core principle: Automate what the camera can reliably know,
let humans record what only humans can know,
and give the owner the complete picture.

Contents

A blueprint for the owner, the clerk, the customer — and the intelligence between them.

Part I — Vision & Users	4
1 Product Vision	4
2 Three Experiences	5
3 Accounts & Access	6
Part II — Clerk Operations	7
4 Clerk Workspace & Shifts	7
Part III — Owner Command Center	8
5 Owner Dashboard & Live Lounge	8
6 TV & Camera Management	9
Part IV — Detection Engine	10
7 The AI Layer	10
8 Game Detection & Counter	10
Part V — Money & Accountability	11
9 Manual Entries & Adjustments	11
10 Payments & Cash	11
Part VI — Operations & Maintenance	12
11 Problems, Incidents & Maintenance	12
12 TV Business Analytics	13
Part VII — Customers & Growth	13
13 Happy Hour	13
14 Customer Experience	14
Part VIII — Tournaments	15
15 Tournaments	15
Part IX — Intelligence & Reports	16
16 Reports & Business Intelligence	16
Part X — Platform & Business	17
17 Settings, Support & Advertising	17
18 Subscription Model	18
19 Data Architecture & Audit	19
Part XI — Principles & Roadmap	20
20 Product Rules We Never Break	20
21 Development Roadmap	21
22 The Final GameWatch Product	22

Product at a Glance

EXPERIENCES 3 user roles	CORE OBJECTS 30 data model	BUILD PHASES 8 roadmap	PRODUCT RULES 10 never broken	PLANS 3 Free/Plus/Pro
---------------------------------------	---	-------------------------------------	--	------------------------------------

PRODUCT	GameWatch
PURPOSE	Game-lounge operating, monitoring, payment, staff, customer, tournament, and business-intelligence platform
PRIMARY MARKET	Game/FIFA/FC lounges, initially Ethiopia, designed for international expansion
CORE PRINCIPLE	Automate what the camera can reliably know, let humans record what only humans can know, and give the owner the complete picture.

The platform in five layers

INTELLIGENCE	Utilization, peak hours, maintenance patterns, detection quality and AI analysis — AI explains, rules decide
CUSTOMER EXPERIENCE	Lounge page, live TV availability, waiting list, Happy Hour, promotions, tournaments
OWNER & CLERK APPS	Owner dashboard, live cameras, shifts, payments, evidence, problems, reports, settings
RECORDS & MONEY	Sessions, games, transactions, adjustments, payments, audit logs — never destroyed by a new day
DETECTION ENGINE	Camera → TV → scoreboard reader → time-aware brain → match verifier → game counter

What is GameWatch?

GameWatch is a smart operating platform for game lounges that uses existing cameras and computer vision to automatically understand game activity, while giving owners tools for staff management, payments, revenue tracking, maintenance, promotions, Happy Hours, tournaments, customer engagement and business intelligence.

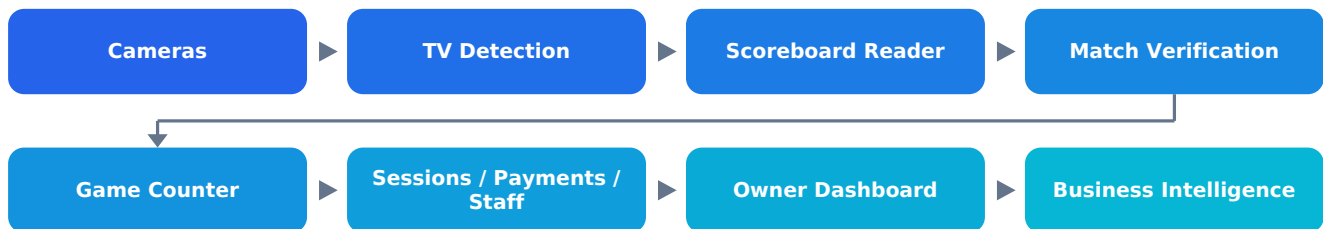
Product Vision

One problem, connected end-to-end

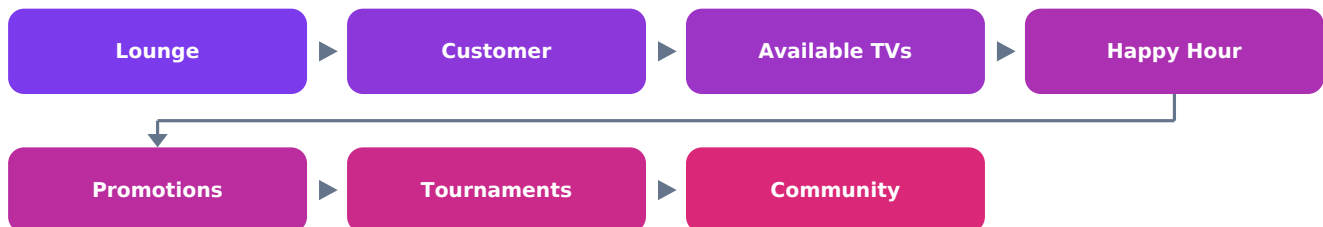
PRD \$1

Game-lounge owners have TVs, customers, clerks, money and cameras, but very little automated understanding of what is actually happening.

GameWatch connects those pieces



Then the customer side builds on top



OWNER

Full access

"Show me my business."

- ✓ monitor cameras
- ✓ manage TVs
- ✓ manage prices
- ✓ manage clerks
- ✓ monitor shifts
- ✓ see revenue
- ✓ see payments
- ✓ see adjustments
- ✓ see complaints
- ✓ see maintenance
- ✓ see historical records
- ✓ see analytics
- ✓ see weekly/monthly reports
- ✓ create Happy Hours
- ✓ create promotions
- ✓ create tournaments
- ✓ manage customers/community settings
- ✓ manage subscriptions
- ✓ manage payment methods
- ✓ manage cameras
- ✓ manage lounge settings

CLERK

Operations

"Help me run today."

- ✓ log in with assigned lounge code + personal PIN
- ✓ use authorized device
- ✓ start shift
- ✓ manage today's sessions
- ✓ add/correct games
- ✓ record payments
- ✓ upload payment evidence
- ✓ report problems
- ✓ report customer complaints
- ✓ report damaged equipment
- ✓ upload photos
- ✓ handle major incidents
- ✓ end shift
- ✓ record cash/deposit information where required

Cannot normally see

- ✗ historical revenue
- ✗ weekly business reports
- ✗ monthly reports
- ✗ owner analytics
- ✗ other clerks' private information
- ✗ subscription
- ✗ lounge settings
- ✗ business intelligence
- ✗ staff management

CUSTOMER

Separate experience

"Help me find a game, know when to play, and have more fun."

- ✓ discover/follow a lounge
- ✓ see lounge information
- ✓ see available TVs
- ✓ see estimated waiting time
- ✓ join a waiting list
- ✓ see Happy Hour
- ✓ see promotions
- ✓ see tournaments
- ✓ join tournaments
- ✓ see tournament schedules
- ✓ see their match time
- ✓ see tournament brackets
- ✓ see results
- ✓ receive lounge notifications

GameWatch Intelligence — "Help everyone understand what is happening."

FIRST SCREEN

Smart game-lounge management

OWNER

CLERK

CUSTOMER

Owner **Creates / logs in**

Clerk **Credentials from owner**

Customer **Connects to lounge**

CREATE LOUNGE

Lounge name

Owner name

Phone / email

Password

Country

Currency

Continue

LOUNGE SETUP

Configure now or later

- ✓ Number of TVs
- ✓ Payment methods
- ✓ Opening hours
- ✓ Default game price
- ✓ Camera setup

The owner can continue configuring the lounge later.

Clerk security model

LOUNGE

Lounge code

+

CLERK

Clerk identity

+

DEVICE

Authorized device

ADD CLERK

Name

Phone

Generate access code

Create personal PIN

EXAMPLE CREDENTIALS

Lounge Code **GW-4827**Clerk PIN **5831**

Not a permanent reusable public invitation.

DEVICE ALERT

⚠ **New device detected**

- ✓ Active devices are recorded
- ✓ Owner can investigate or revoke access
- ✓ Activity / login audit trail maintained

Clerk Workspace & Shifts

Simple by design — a fresh workspace every business day

PRD §8-12

TODAY

LIVE TVs

MANAGE GAMES

PAYMENTS

REPORT PROBLEM

MY SHIFT

GOOD MORNING, HANA

Your previous shift:

Yesterday **17:02 - 23:14**

Today's workspace is ready.

START SHIFT

SHIFT STARTED

08:57 AM

Shift status: active

Games

0

Sessions

0

Adjustments

0

Problems

0

● clerk

● date / business day

● start time

● device

● shift status

GameWatch records each of these when the shift starts.

Business day — not just midnight

MIDNIGHT — not the end of the working day

OCTOBER 2 BUSINESS DAY · 10:00 AM → 2:00 AM

10:00 AM

OCTOBER 3 · fresh workspace

2:00 AM starts per the lounge's configured schedule

OCTOBER 3

Blank next-day workspace

Games

0

Sessions

0

Adjustments

0

Problems

0

START SHIFT

END SHIFT

Started **08:57**

Games handled **48**

Adjustments **3**

Problems **2**

Cash **800 ETB**

Telebirr **400 ETB**

Total **1,200 ETB**

END SHIFT

NEVER DELETED

- ✓ Yesterday's data is not deleted
- ✓ It simply isn't part of the clerk's normal daily workspace
- ✓ The owner retains the historical records
- ✓ Exact end time and the state of the lounge/work are recorded

Owner Dashboard & Live Lounge

The command center — and full transparency into what the system sees

PRD §13-14

Today

REVENUE

1,425 ETB

GAMES

57

ACTIVE TVS

5 / 8

CLERKS

2

Live lounge

● full camera feeds

● detected TVs

● cropped TV regions

● current match state

● clock

● detection confidence

● TV status

The diagram illustrates the system's ability to switch from a full camera view to a focused TV crop view. On the left, the 'CAMERA 1 · FULL CAMERA VIEW' shows a dark background with two blue rectangular regions representing detected TVs. The first region is labeled 'TV 1 · 94%' and the second is 'TV 2 · 91%'. An arrow points from this view to the right, where the 'TV 1 · TV CROP VIEW' is shown. This view displays a large digital clock showing '37:42' with the text 'cropped scoreboard / TV area' below it. To the right of the clock, the 'Clock' is listed as '37:42', the 'Status' is 'ACTIVE' (in a green pill), and the 'Confidence' is '94%' with a corresponding green progress bar.

Left: the complete camera image with detected TV regions. Right: switch to the TV crop view.

This gives the owner transparency into what the system actually sees.

ADD TV

TV name

Location

Game

Price per game

Connected camera

Detected region

SAVE

EACH TV HAS ITS OWN

- ✓ name
- ✓ location
- ✓ price
- ✓ camera
- ✓ detection region
- ✓ game
- ✓ status
- ✓ utilization
- ✓ revenue
- ✓ maintenance history

SUPPORTED CAMERAS

- ✓ phone camera
- ✓ PC / webcam
- ✓ USB camera
- ✓ security camera
- ✓ CCTV / IP camera
- ✓ compatible network video stream

One camera may monitor multiple TV regions.

Normalized into one internal video interface

Camera

Camera Adapter

GameWatch Video Stream

TV Detection

Scoreboard Reader

Camera health

● Healthy

● Poor detection

● Disconnected

● Obstructed

● Too blurry

● Too dark

CAMERA 3

⚠ TV 4 detection confidence has been unusually low.

RULE 7

Camera failure must be visible rather than silently producing fake data.

The AI Layer

AI explains. Deterministic GameWatch rules decide.

PRD §18

CORE GAMEWATCH — DECIDES

"I detected this game."

- ✓ Deterministic rules
- ✓ Authority for financial truth
- ✓ Records the game

ANTIGRAVITY — ANALYSES

"Why is detection failing?"

- ✓ Analysis layer only
- ✓ Never the authority for financial decisions
- ✓ Explains patterns

EXAMPLE: TV 4

Normally 92-95% **92-95%**

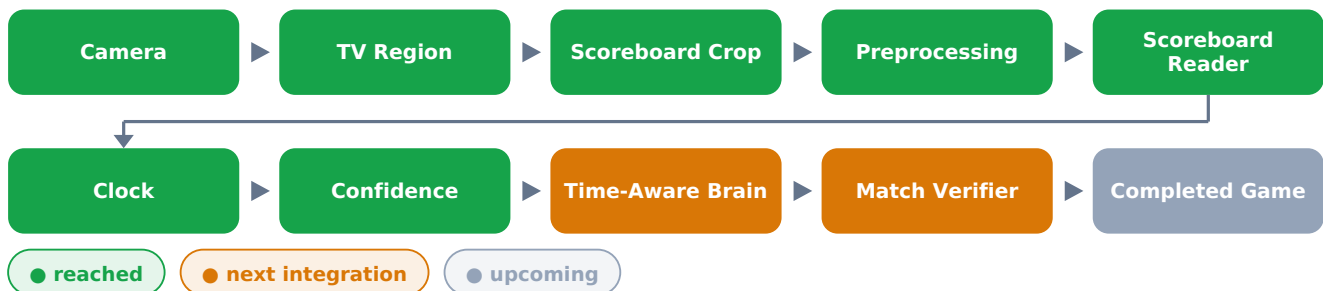
Today **61-70%**

AI analysis: camera position appears to have changed or the scoreboard region is partially obscured.

Game Detection & Counter

From camera pixels to a completed, priced game

PRD §19-20



GAME COUNTER RECORD

TV	TV 3
Event	Game completed
Price	25 ETB
Time	18:43
Detection confidence	94%

The game automatically becomes part of the lounge's records.

PHASE 1 STATUS

Scoreboard reader

Implemented and tested against current captured examples.

- ✓ Successful readings: 00:51, 01:21, 03:15
- ✓ Confidence roughly 89-94%
- ✓ Next: time-aware brain + match verifier

Manual Entries & Adjustments

A clerk can never simply erase revenue

PRD §21-23

ADD GAME

TV

Game

Price

Why added manually?

- ☒ GameWatch failed to detect it
- ☐ Camera unavailable
- ☐ Customer arrangement
- ☐ Loyal customer/package
- ☐ Other

If needed: explain what happened.

ADJUST TRANSACTION

Original

25 ETB

Reason

- ☐ Customer did not finish
- ☒ Technical problem
- ☐ Detection error
- ☐ Loyal customer
- ☐ Package
- ☐ Other

Explanation

SUBMIT

WHY IT MATTERS

- ✓ Every adjustment is audited
- ✓ Reasons are required
- ✓ Shows why automatic detection isn't matching reality
- ✓ Owner can analyze discounts later

Customer discount example

20 GAMES × 25 ETB

500 ETB

—

DISCOUNT

100 ETB

=

FINAL PAYMENT

400 ETB

Reason recorded: loyal customer / package.

Payments & Cash

Configurable methods, attached evidence, reconciled cash

PRD §24-26

Cash

Telebirr

CBE

Example Ethiopia setup — each lounge configures its own supported payment methods.

PAYMENT EVIDENCE

Method

Telebirr

Amount

400 ETB

+ Upload payment proof

SAVE

CASH SETTLEMENT

Expected cash

800 ETB

Actual cash

780 ETB

Difference -20 ETB

Reason

SUBMIT

OR REQUIRE A DEPOSIT

Amount

Bank

+ Receipt photo

SUBMIT

The payment screenshot/photo becomes attached to the transaction — same concept for CBE. The owner sees the record.

PART VI — OPERATIONS & MAINTENANCE

11

Problems, Incidents & Maintenance

Every issue recorded, tracked, and remembered

PRD §27-29

REPORT PROBLEM

Type

- ☒ Controller damaged
- ☐ Controller missing
- ☐ TV problem
- ☐ Console problem
- ☐ Internet problem
- ☐ Payment problem
- ☐ Camera problem
- ☐ Customer complaint
- ☐ Other

TV

TV 3

Description

+ Add photo

Normal

Important

Major

SUBMIT

OWNER ALERT

▲ MAJOR ISSUE

TV

TV 3

Issue **Controller damaged**

Reported by **Hana**

Time **18:43**

Photo attached

INVESTIGATE

MAJOR INCIDENTS

CALL OWNER

For major issues the clerk can call the owner while the incident is also recorded.

Status flow

OPEN

INVESTIGATING

FIXED

CLOSED

Incident lifecycle

OPEN

INVESTIGATING

FIXED

CLOSED

Maintenance intelligence

TV 3 · Controller reports

Sept 21

Sept 25

Sept 29

Repeated controller-related problems have been recorded for TV 3.

TV 1 · BUSINESS PROFILE

Games

24

Revenue

600 ETB

Utilization

41%

Detection

Good

Problems

4

Customer reports

3

NOT A VERDICT

GameWatch should not simply declare:

"TV 1 is bad."

Instead it separates:

- > Observed
- > Related recorded data
- > Possible explanations

OBSERVED

TV 1 had lower utilization than the lounge average.

RELATED RECORDED DATA

- lower game count
- controller complaints
- availability periods
- camera problems
- location
- customer reports

POSSIBLE EXPLANATIONS

- ? Controller issues may be contributing.
- ? TV location may be affecting usage.
- ? Customers may prefer other TVs.

These are analysis / hypotheses — not fabricated facts. The owner can investigate underperforming TVs.

PART VII — CUSTOMERS & GROWTH

NORMAL PRICE

25 ETB



HAPPY HOUR

15 ETB

OWNER: HAPPY HOUR

Days **Monday-Thursday**
Time **2 PM - 5 PM**
Normal price **25 ETB**
Happy Hour **15 ETB**

ACTIVATE

CUSTOMER VIEW

HAPPY HOUR

15 ETB / GAME

2 PM - 5 PM

Ends in

01:24

VIEW LOUNGE

SMART SUGGESTION

GameWatch can identify consistently low-utilization periods and suggest Happy Hour windows.

The owner makes the final decision.

LIVE AVAILABILITY

TV 1	Playing
TV 2	Available
TV 3	Available
TV 4	Ending soon

Customers can avoid unnecessary waiting.

ALL TVs BUSY

Estimated wait:

18 min

You are #3 in line

JOIN WAITING LIST

● TV 4 is available.

CHAMPION GAME LOUNGE

8 TVs · FIFA / FC

Availability

3 TVs free

Happy Hour

Tournament

Promotions

Follow Lounge

View TVs

Tournaments

Lounge promotions

New TVs have arrived

Cold drinks available

More space available

New PS5s installed

Tournament this Saturday

Happy Hour 2-5 PM

Owner can attach an image. Customers following the lounge receive the promotion.

20 GAME PACKAGE

Normal	500 ETB
Package	450 ETB
Games remaining	14 left

Avoids clerks manually inventing discounts every time.

PROMOTION

New TVs have arrived

+ Promotion image

Delivered to customers following the lounge.

Tournaments

Create, register, bracket, play, record

PRD §38-43

CREATE TOURNAMENT

Name

Date

Start time

Game

Maximum players

Entry fee

Prize

Format

CREATE

FRIDAY NIGHT CHAMPIONSHIP

32 players

Entry

200 ETB

Winner

1,000 ETB

Spots filled

16 / 32

JOIN

MY TOURNAMENT

Quarter Final

Opponent

Messay

TV

TV 4

Estimated time

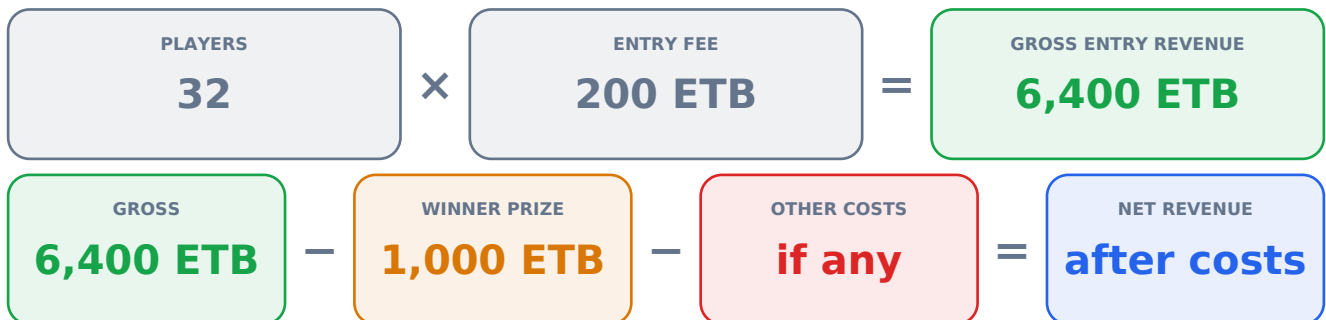
7:40 PM

VIEW BRACKET

You won 3-1.

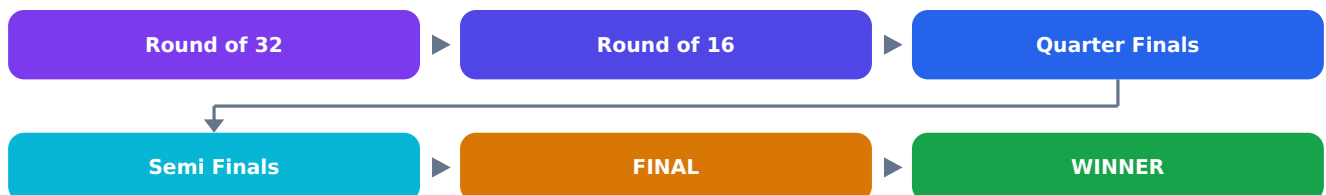
You lost 3-2.

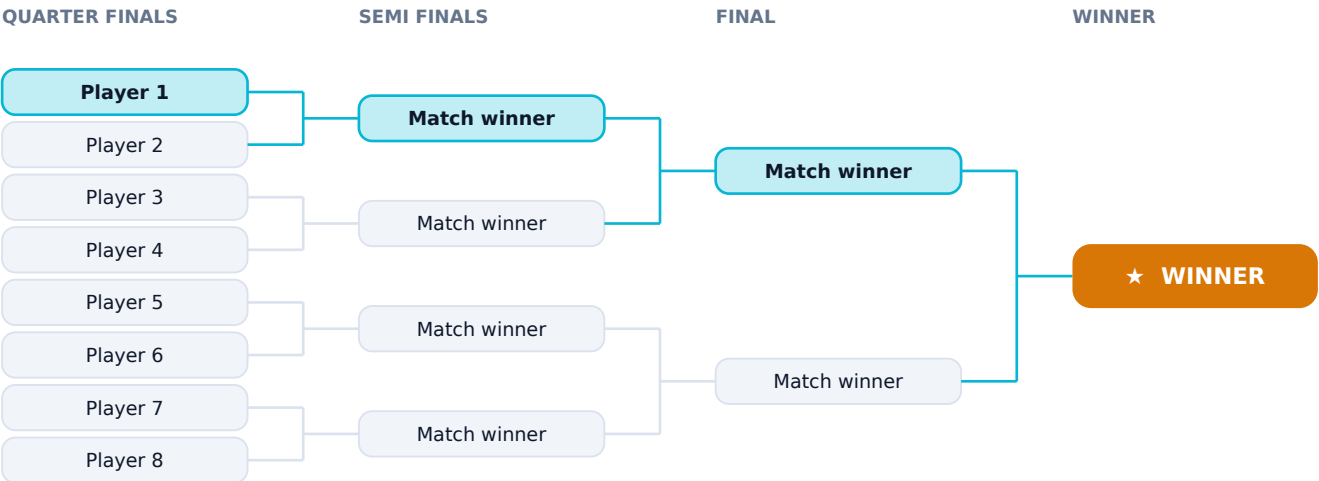
Tournament economics



Other costs, if any, are separately recorded before calculating net tournament revenue.

Bracket maintained automatically





Every match result is recorded and updates the bracket.

TOURNAMENT WAITING

Your match is estimated for 8:15 PM.

Your match is coming up.

The customer doesn't need to sit beside the TV all day.

OWNER DASHBOARD

Players	32 / 32
Entry revenue	6,400 ETB
Prize	1,000 ETB
Matches	24 / 31
Current round	Quarter Final

PART IX — INTELLIGENCE & REPORTS

16

Reports & Business Intelligence

Data → Analysis → Possible Explanation

PRD \$44-45

Weekly report

GAMES

343

REVENUE

8,575 ETB

AVG GAMES / DAY

49

AVG REVENUE / DAY

1,225 ETB

TV performance

Peak hours

Payment breakdown

Discounts

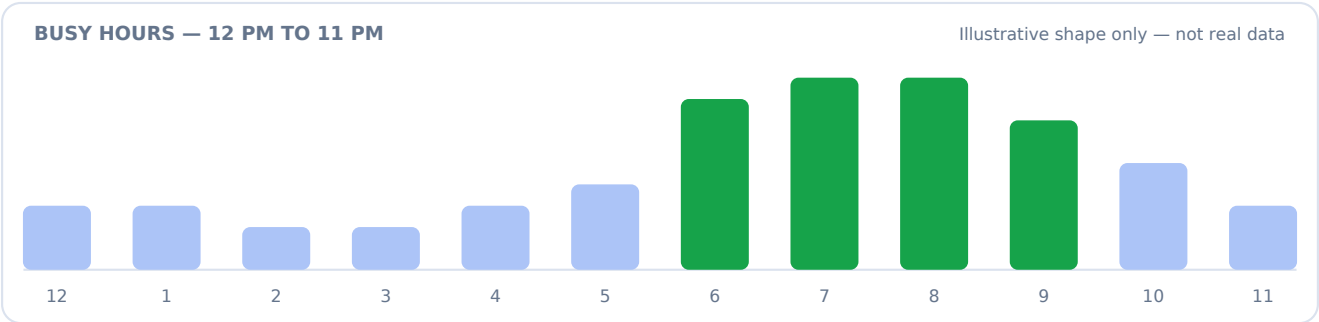
Adjustments

Clerk activity

Maintenance

Detection failures

Tournament performance



Highlighted: 6 PM-9 PM is the busiest period.

GameWatch Intelligence — this week

REVENUE

↑ **12%**
vs previous week

GAMES

↑ **8%**
vs previous week

- ⚠ TV 1 utilization is low.
- ⚠ TV 3 had repeated controller reports.
- Camera 3 had poor detection confidence several times.
- ↑ 6 PM-9 PM is the busiest period.
- ★ Thursday 2 PM-5 PM has consistently low utilization.

The system always distinguishes

DATA

ANALYSIS

POSSIBLE EXPLANATION

PART X — PLATFORM & BUSINESS

17

Settings, Support & Advertising

Configurable per lounge — international from day one

PRD \$47-50

LOUNGE

- Name
- Logo
- Country
- Currency
- Address
- Opening hours

TVs

- Add
- Remove
- Rename
- Price
- Game
- Location

CAMERAS

- Add
- Remove
- Camera type
- Connection
- TV regions
- Health

PAYMENTS

- Add payment method
- QR code
- Bank information

STAFF

- Add clerk
- Disable clerk
- Generate credentials
- Devices
- Activity

NOTIFICATIONS

- Camera problems
- Major incidents
- Revenue discrepancies
- Device login alerts

SECURITY

- Password
- PIN
- Active devices
- Login history

SUBSCRIPTION

- Current plan
- Trial
- Renewal
- Upgrade

INTERNATIONALIZATION

- ✓ Country
- ✓ Currency
- ✓ Language
- ✓ Payment methods

Never hard-code ETB.

Ethiopia: 25 ETB. Other countries configure their own currency and pricing.

CONTACT US

Phone **GameWatch support**

Email **GameWatch support**

Contact Support

Report Technical Problem

Reports can include screenshots/photos and relevant system information.

ADVERTISEMENT

Local gaming store

Controllers • Consoles • Accessories

Want your business advertised here?

Contact GameWatch

An additional GameWatch revenue stream.

18

Subscription Model

Free to start, Plus to operate, Pro to grow

PRD §51-54

FREE

1 TV

Basic everything

- ✓ Basic dashboard
- ✓ Basic detection
- ✓ Basic game tracking
- ✓ Basic history

Does not include

- ✗ Clerk access
- ✗ Multiple TVs
- ✗ Advanced analytics
- ✗ Advanced camera management
- ✗ Tournament system
- ✗ Advanced customer features

PLUS

Growing lounges

- ✓ Multiple TVs
- ✓ Multiple cameras
- ✓ Clerk accounts
- ✓ Shift management
- ✓ Payments
- ✓ Payment evidence
- ✓ Daily/weekly reports
- ✓ Problems/maintenance
- ✓ Customer lounge
- ✓ Happy Hour
- ✓ Promotions
- ✓ Basic analytics

PRO

Larger lounges

Everything in Plus, plus:

- ✓ Advanced business intelligence
- ✓ AI analysis
- ✓ Advanced camera monitoring
- ✓ Tournament system
- ✓ Customer community
- ✓ Tournament brackets
- ✓ Advanced reports
- ✓ TV utilization intelligence
- ✓ Maintenance intelligence
- ✓ Advanced customer engagement
- ✓ Advanced promotions

30-day free trial

Sign up + enter payment info

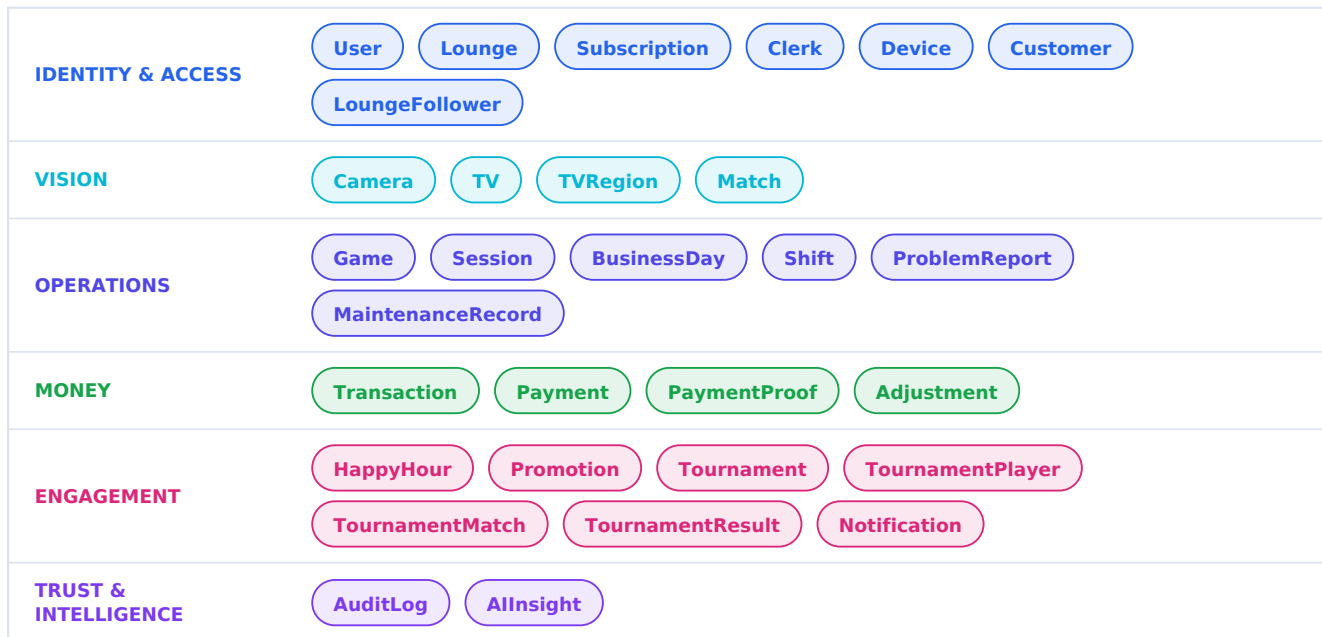
30-day free trial

Choose 3 months · 6 months · 1 year

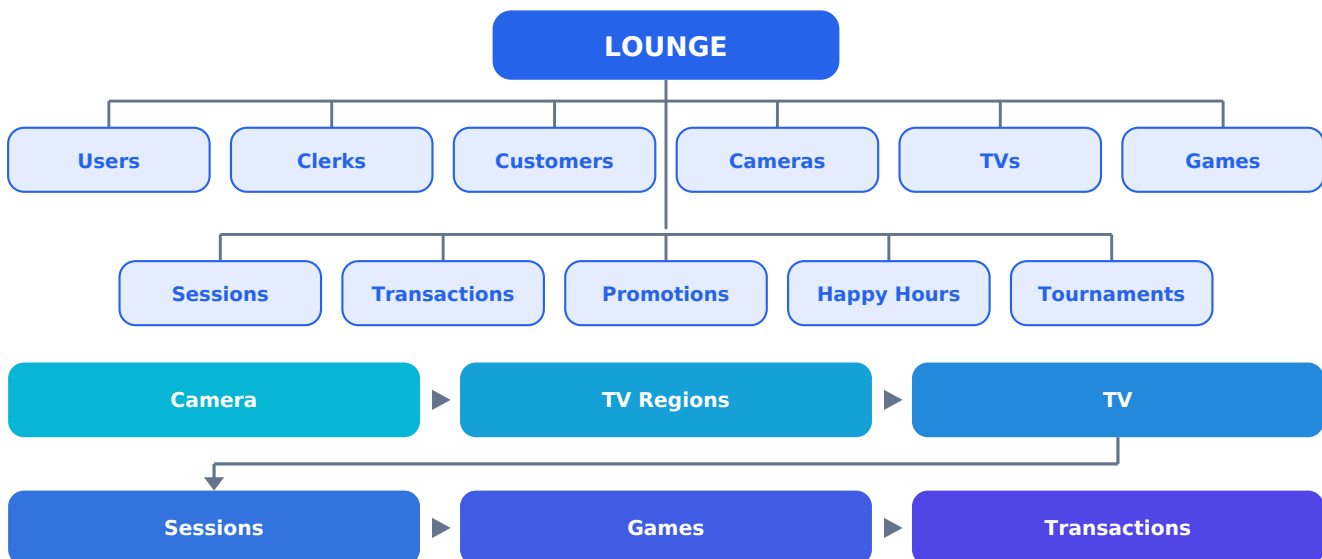
Charged unless cancelled

30-day free trial. You will be charged after the trial according to the selected plan unless you cancel.

Pricing is configured separately.



Most important relationships



Audit log — everywhere money or permissions are involved

AUDIT ENTRY · EXAMPLE		WHAT IS RECORDED	
Time	18:42	✓ who did it	
Who	Hana	✓ what happened	
Record	Transaction #1042	✓ when	
Change	25 ETB → 0 ETB	✓ from which device	
Reason	Technical problem	✓ previous value	
Device	Samsung A03	✓ new value	
		✓ reason	
		Extremely important for business trust.	

Product Rules We Never Break

Ten principles to check every decision against

PRD §58

1

AI does not decide financial truth.

6

Customer cannot access private lounge business information.

2

Clerk adjustments require reasons.

7

Camera failure must be visible rather than silently producing fake data.

3

Historical records are never destroyed merely because a new day starts.

8

GameWatch should never pretend it knows something it did not observe.

4

Owner has full visibility.

9

Currency and payment methods are configurable by lounge.

5

Clerk sees only what she needs to operate.

10

Every important financial/security action is auditable.

WE ARE HERE



Phase 1 status: the scoreboard reader is implemented and tested (readings 00:51, 01:21, 03:15 at roughly 89-94% confidence). Next: time-aware brain, match verifier, game counter.

GameWatch is a smart operating platform for game lounges that uses existing cameras and computer vision to automatically understand game activity, while giving owners tools for staff management, payments, revenue tracking, maintenance, promotions, Happy Hours, tournaments, customer engagement and business intelligence.

OWNER

"Show me my business."

CLERK

"Help me run today."

CUSTOMER

"Help me find a game, know when to play, and have more fun."

INTELLIGENCE

"Help everyone understand what is happening."

This is the GameWatch PRD v1.0. When we build something, we check it against this blueprint instead of randomly changing the product architecture.